

# ENTREGA FINAL AWS ARCHITECTURE

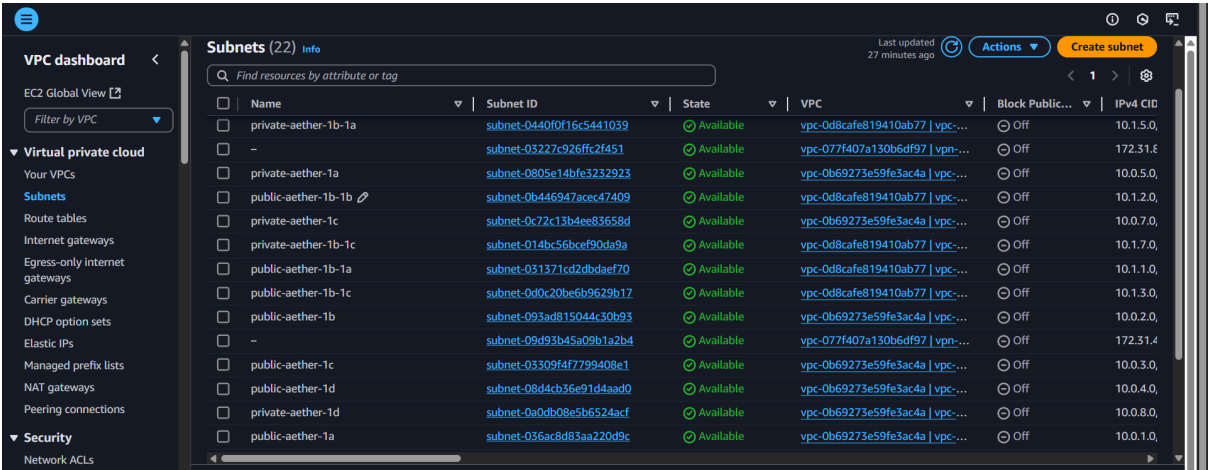
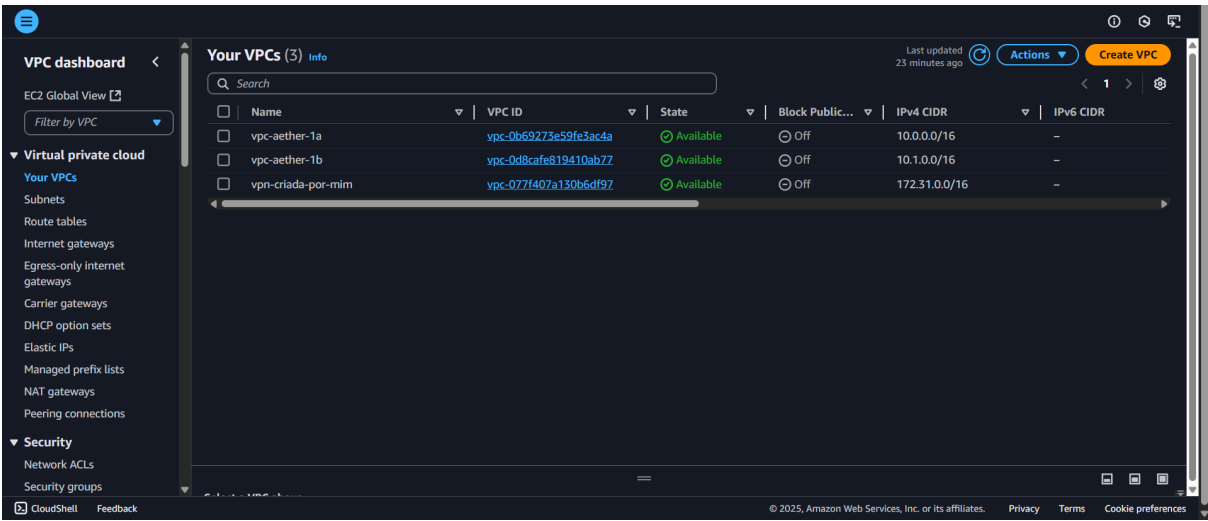
## (Tema: SENAI Tank Brasil)

Grupo: Antônio Cerdeira, João Paulo Tombi, Luane Patricio, Lucas Romão, Lorena Gabarão e Samuel Canuto

**Aether é uma empresa de soluções em nuvem baseada nos serviços da AWS.** Apresentamos nosso próximo case de sucesso, a infraestrutura criada para a empresa de varejo Finexo, que prevê receber mais de 2 milhões de acessos na próxima Black Friday.

**VPC** é a sigla para "Virtual Private Cloud" ou "Nuvem Privada Virtual". É uma rede virtual que pode ser hospedada em uma nuvem pública. Integrado a VPC, criamos também:

- **Subnets**
- **Route Tables**
- **Internet Gateway**
- **NAT gateway**



|                      |                          |                      |                                          |           |                                       |
|----------------------|--------------------------|----------------------|------------------------------------------|-----------|---------------------------------------|
| DHCP option sets     | <input type="checkbox"/> | -                    | <a href="#">subnet-0340aca4e944eb787</a> | Available | <a href="#">vpc-077f407a130b6df97</a> |
| Elastic IPs          | <input type="checkbox"/> | private-aether-1b-1d | <a href="#">subnet-0ffa4b6a0c06ace3f</a> | Available | <a href="#">vpc-0d8cafe819410ab77</a> |
| Managed prefix lists | <input type="checkbox"/> | private-aether-1b-1b | <a href="#">subnet-0b278df4f55da1c8b</a> | Available | <a href="#">vpc-0d8cafe819410ab77</a> |
| NAT gateways         | <input type="checkbox"/> | public-aether-1b-1d  | <a href="#">subnet-020ef3312790948dd</a> | Available | <a href="#">vpc-0d8cafe819410ab77</a> |
| Peering connections  | <input type="checkbox"/> | private-aether-1b    | <a href="#">subnet-0c08f4d8dd3275a5c</a> | Available | <a href="#">vpc-0b69273e59fe3ac4a</a> |

|                               |                                                                                                                                           |  |  |  |  |  |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| <b>VPC dashboard</b>          | <b>Route tables (7)</b> <small>Info</small> <span>Last updated 27 minutes ago</span> <span>Actions</span> <span>Create route table</span> |  |  |  |  |  |
| EC2 Global View               | Find resources by attribute or tag                                                                                                        |  |  |  |  |  |
| Filter by VPC                 |                                                                                                                                           |  |  |  |  |  |
| <b>Virtual private cloud</b>  |                                                                                                                                           |  |  |  |  |  |
| Your VPCs                     |                                                                                                                                           |  |  |  |  |  |
| Subnets                       |                                                                                                                                           |  |  |  |  |  |
| Route tables                  |                                                                                                                                           |  |  |  |  |  |
| Internet gateways             |                                                                                                                                           |  |  |  |  |  |
| Egress-only internet gateways |                                                                                                                                           |  |  |  |  |  |
| Carrier gateways              |                                                                                                                                           |  |  |  |  |  |
| DHCP option sets              |                                                                                                                                           |  |  |  |  |  |
| Elastic IPs                   |                                                                                                                                           |  |  |  |  |  |

| <input type="checkbox"/> | Name                 | Route table ID                        | Explicit subnet associ... | Edge associations | Main | VPC                                   |
|--------------------------|----------------------|---------------------------------------|---------------------------|-------------------|------|---------------------------------------|
| <input type="checkbox"/> | -                    | <a href="#">rtb-0b9986e2d4c5c183c</a> | -                         | -                 | Yes  | <a href="#">vpc-0d8cafe819410ab77</a> |
| <input type="checkbox"/> | public-rt-aether-1a  | <a href="#">rtb-03d2b9c8cfc8c6fc</a>  | 4 subnets                 | -                 | No   | <a href="#">vpc-0b69273e59fe3ac4a</a> |
| <input type="checkbox"/> | private-rt-aether-1a | <a href="#">rtb-0beb65d3461d6fe5e</a> | 4 subnets                 | -                 | No   | <a href="#">vpc-0b69273e59fe3ac4a</a> |
| <input type="checkbox"/> | -                    | <a href="#">rtb-02d3748eda03b1615</a> | 6 subnets                 | -                 | Yes  | <a href="#">vpc-077f407a130b6df97</a> |
| <input type="checkbox"/> | private-rt-aether-1b | <a href="#">rtb-0dfe486ce010ce133</a> | 4 subnets                 | -                 | No   | <a href="#">vpc-0d8cafe819410ab77</a> |
| <input type="checkbox"/> | public-rt-aether-1b  | <a href="#">rtb-01599146abfa092fa</a> | 4 subnets                 | -                 | No   | <a href="#">vpc-0d8cafe819410ab77</a> |
| <input type="checkbox"/> | -                    | <a href="#">rtb-06d998aa057edfadb</a> | -                         | -                 | Yes  | <a href="#">vpc-0b69273e59fe3ac4a</a> |

|                               |                                                                                                            |  |  |  |  |  |
|-------------------------------|------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| <b>VPC dashboard</b>          | <b>Internet gateways (3)</b> <small>Info</small> <span>Actions</span> <span>Create internet gateway</span> |  |  |  |  |  |
| EC2 Global View               | Search                                                                                                     |  |  |  |  |  |
| Filter by VPC                 |                                                                                                            |  |  |  |  |  |
| <b>Virtual private cloud</b>  |                                                                                                            |  |  |  |  |  |
| Your VPCs                     |                                                                                                            |  |  |  |  |  |
| Subnets                       |                                                                                                            |  |  |  |  |  |
| Route tables                  |                                                                                                            |  |  |  |  |  |
| Internet gateways             |                                                                                                            |  |  |  |  |  |
| Egress-only internet gateways |                                                                                                            |  |  |  |  |  |
| Carrier gateways              |                                                                                                            |  |  |  |  |  |

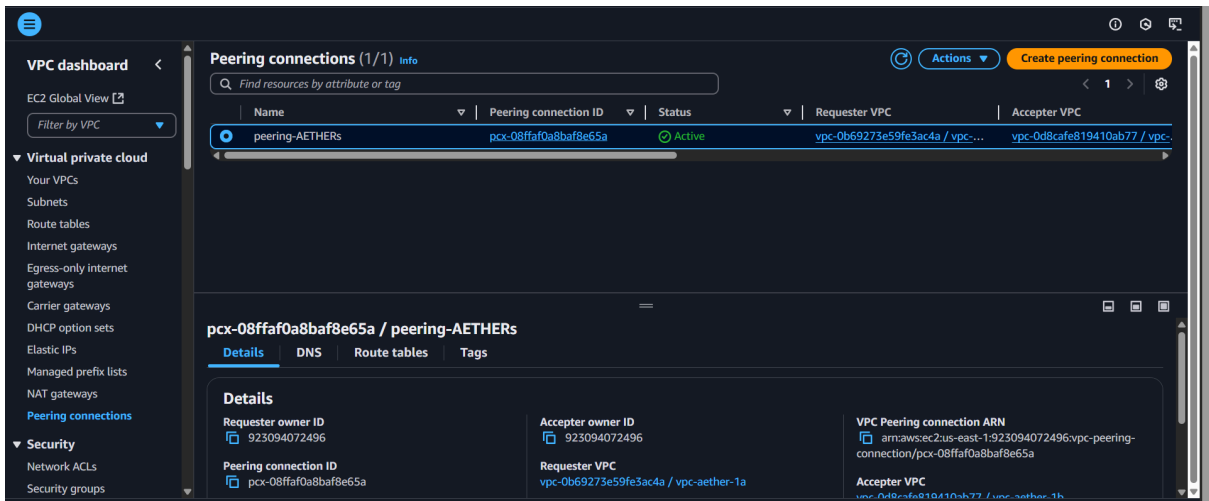
| <input type="checkbox"/> | Name          | Internet gateway ID                   | State    | VPC ID                                                                  | Owner        |
|--------------------------|---------------|---------------------------------------|----------|-------------------------------------------------------------------------|--------------|
| <input type="checkbox"/> | -             | <a href="#">igw-01031ea4a569d0bf4</a> | Attached | <a href="#">vpc-077f407a130b6df97</a>   <a href="#">vpn-criada-p...</a> | 923094072496 |
| <input type="checkbox"/> | aether-1b-igw | <a href="#">igw-021042274ac75d78c</a> | Attached | <a href="#">vpc-0d8cafe819410ab77</a>   <a href="#">vpc-aether-1b</a>   | 923094072496 |
| <input type="checkbox"/> | aether-igw    | <a href="#">igw-04f7a6fc2c9394703</a> | Attached | <a href="#">vpc-0b69273e59fe3ac4a</a>   <a href="#">vpc-aether-1a</a>   | 923094072496 |

|                               |                                                                                                 |  |  |  |  |  |
|-------------------------------|-------------------------------------------------------------------------------------------------|--|--|--|--|--|
| <b>VPC dashboard</b>          | <b>Elastic IP addresses (2/3)</b> <span>Actions</span> <span>Allocate Elastic IP address</span> |  |  |  |  |  |
| EC2 Global View               | Find resources by attribute or tag                                                              |  |  |  |  |  |
| Filter by VPC                 |                                                                                                 |  |  |  |  |  |
| <b>Virtual private cloud</b>  |                                                                                                 |  |  |  |  |  |
| Your VPCs                     |                                                                                                 |  |  |  |  |  |
| Subnets                       |                                                                                                 |  |  |  |  |  |
| Route tables                  |                                                                                                 |  |  |  |  |  |
| Internet gateways             |                                                                                                 |  |  |  |  |  |
| Egress-only internet gateways |                                                                                                 |  |  |  |  |  |
| Carrier gateways              |                                                                                                 |  |  |  |  |  |

| <input checked="" type="checkbox"/> | Name | Allocated IPv4 addr...        | Type      | Allocation ID                              | Reverse DNS record |
|-------------------------------------|------|-------------------------------|-----------|--------------------------------------------|--------------------|
| <input checked="" type="checkbox"/> | -    | <a href="#">18.215.13.218</a> | Public IP | <a href="#">eipalloc-047f588a64c6fdd66</a> | -                  |
| <input type="checkbox"/>            | -    | <a href="#">54.163.135.62</a> | Public IP | <a href="#">eipalloc-0db1431e872ef7bb8</a> | -                  |
| <input checked="" type="checkbox"/> | -    | <a href="#">54.221.88.240</a> | Public IP | <a href="#">eipalloc-0ae84c73a416e09cb</a> | -                  |

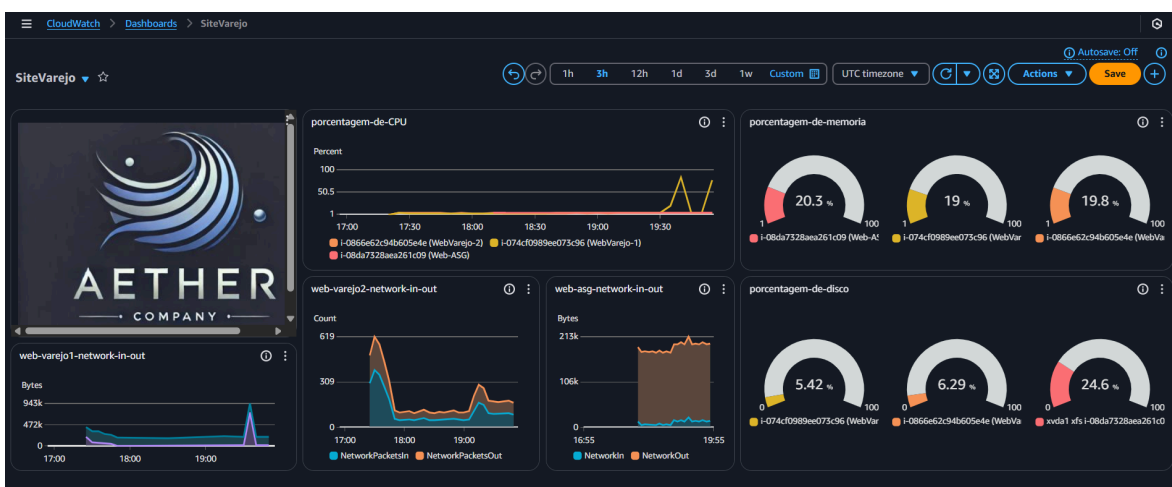
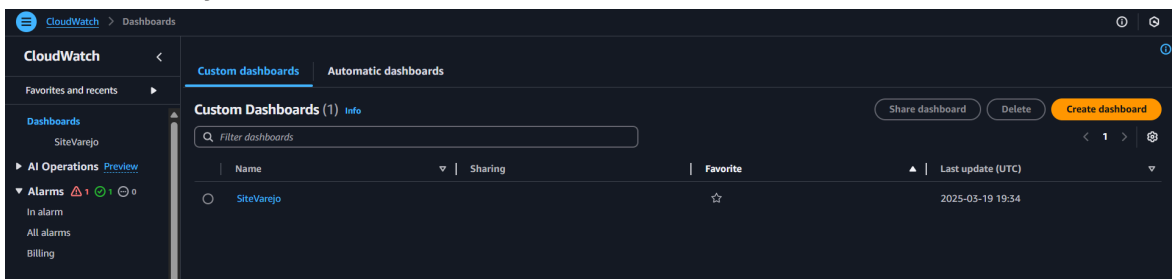
|                               |                                                                                                  |  |  |  |  |  |
|-------------------------------|--------------------------------------------------------------------------------------------------|--|--|--|--|--|
| <b>VPC dashboard</b>          | <b>NAT gateways (3)</b> <small>Info</small> <span>Actions</span> <span>Create NAT gateway</span> |  |  |  |  |  |
| EC2 Global View               | Find resources by attribute or tag                                                               |  |  |  |  |  |
| Filter by VPC                 |                                                                                                  |  |  |  |  |  |
| <b>Virtual private cloud</b>  |                                                                                                  |  |  |  |  |  |
| Your VPCs                     |                                                                                                  |  |  |  |  |  |
| Subnets                       |                                                                                                  |  |  |  |  |  |
| Route tables                  |                                                                                                  |  |  |  |  |  |
| Internet gateways             |                                                                                                  |  |  |  |  |  |
| Egress-only internet gateways |                                                                                                  |  |  |  |  |  |
| Carrier gateways              |                                                                                                  |  |  |  |  |  |

| <input type="radio"/> | Name          | NAT gateway ID                        | Connectivity... | State     | State message | Primary public I...           | Primary priva |
|-----------------------|---------------|---------------------------------------|-----------------|-----------|---------------|-------------------------------|---------------|
| <input type="radio"/> | aether-NAT    | <a href="#">nat-0898b5e3fb34f0e0d</a> | Public          | Available | -             | <a href="#">54.221.88.240</a> | 10.0.5.82     |
| <input type="radio"/> | aether-1b-NAT | <a href="#">nat-07f54e58339d11949</a> | Public          | Available | -             | <a href="#">18.215.13.218</a> | 10.1.1.175    |
| <input type="radio"/> | aether-1b-NAT | <a href="#">nat-07fe2fd478f3be743</a> | Public          | Deleted   | -             | <a href="#">54.163.135.62</a> | 10.0.9.209    |



O **VPC Peering** é uma conexão de rede que permite que duas Virtual Private Clouds (VPCs) se comuniquem diretamente como se estivessem na mesma rede. Ele é usado para conectar VPCs dentro de um mesmo provedor de nuvem (como AWS, Google Cloud ou Azure) sem precisar passar por um gateway de internet, VPN ou outra solução de roteamento.

**CloudWatch:** É um serviço da AWS usado para monitoramento e observabilidade de recursos e aplicações em nuvem. Ele coleta e analisa métricas, logs e eventos de diferentes serviços da AWS e até mesmo de servidores on-premises ou outras nuvens.



Prints das configurações utilizadas na EC2

```
aws [Alt+S] United States (N. Virginia) voclabs/user3922609=luucas.munhoz@aluno.senai.br @ 2382-5678-3131

[root@ip-10-0-5-103 ec2-user]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2025-03-19 18:44:23 UTC; 6s ago
     Docs: man:httpd.service(8)
   Main PID: 6736 (httpd)
   Status: "Started, listening on: port 80"
     Tasks: 177 (limit: 1111)
    Memory: 12.9M
       CPU: 57ms
   CGroup: /system.slice/httpd.service
           └─6736 /usr/sbin/httpd -DFOREGROUND
             └─6737 /usr/sbin/httpd -DFOREGROUND
               └─6738 /usr/sbin/httpd -DFOREGROUND
                 └─6739 /usr/sbin/httpd -DFOREGROUND
                   └─6740 /usr/sbin/httpd -DFOREGROUND

Mar 19 18:44:23 ip-10-0-5-103.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Mar 19 18:44:23 ip-10-0-5-103.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Mar 19 18:44:23 ip-10-0-5-103.ec2.internal httpd[6736]: Server configured, listening on: port 80
[root@ip-10-0-5-103 ec2-user]# systemctl status amazon-cloudwatch-agent
● amazon-cloudwatch-agent.service - Amazon CloudWatch Agent
   Loaded: loaded (/etc/systemd/system/amazon-cloudwatch-agent.service; disabled; preset: disabled)
   Active: active (running) since Wed 2025-03-19 18:44:02 UTC; 36s ago
     Main PID: 6651 (amazon-cloudwat)
       Tasks: 6 (limit: 1111)
    Memory: 234.7M
       CPU: 340ms
   CGroup: /system.slice/amazon-cloudwatch-agent.service
           └─6651 /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -config /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.toml -envconfig

Mar 19 18:44:03 ip-10-0-5-103.ec2.internal start-amazon-cloudwatch-agent[6656]: D! [EC2] Found active network interface
Mar 19 18:44:03 ip-10-0-5-103.ec2.internal start-amazon-cloudwatch-agent[6656]: I! imds retry client will retry 1 times! Detected the instance is EC2

[CloudShell Feedback] © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences
```

(configuração sobre os serviços)

**Load Balance:** Distribui automaticamente o tráfego de rede entre vários servidores para evitar sobrecarga e garantir alta disponibilidade e desempenho.

EC2 > Load balancers > aether-lb

Elastic Block Store

Network & Security

Load Balancing

Auto Scaling

aether-lb

Details

Load balancer type

Scheme

Status

Hosted zone

VPC

Availability Zones

Load balancer IP address type

Date created

Load balancer ARN

DNS name info

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Target Group

EC2 > Target groups > Varejo

Elastic Block Store

Network & Security

Load Balancing

Auto Scaling

Varejo

Details

Target type

IP address type

Protocol : Port

Load balancer

Protocol version

VPC

3 Total targets

0 Anomalous

0 Unhealthy

0 Unused

0 Initial

0 Draining

Distribution of targets by Availability Zone (AZ)

Registered targets (3)

TargetsMonitoringHealth checksAttributesTags

Registered targets (3) Info

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

< 1 >

|                          | Instance ID                         | Name        | Port | Zone              | Health status | Health status details | Admini...    | Overr |
|--------------------------|-------------------------------------|-------------|------|-------------------|---------------|-----------------------|--------------|-------|
| <input type="checkbox"/> | <a href="#">i-08da7328aea261c09</a> | Web-ASG     | 80   | us-east-1d (us... | Healthy       | -                     | No override. | No ov |
| <input type="checkbox"/> | <a href="#">i-074cf0989ee073c96</a> | WebVarejo-1 | 80   | us-east-1a (us... | Healthy       | -                     | No override. | No ov |
| <input type="checkbox"/> | <a href="#">i-0866e62c94b605e4e</a> | WebVarejo-2 | 80   | us-east-1b (us... | Healthy       | -                     | No override. | No ov |

**AutoScaling:** Ajusta automaticamente a quantidade de instâncias (servidores) com base na demanda, aumentando ou reduzindo recursos para otimizar desempenho e custos.

EC2 > Auto Scaling groups > ASG-AETHER1

Elastic Block Store

Network & Security

Load Balancing

Auto Scaling

Settings

ASG-AETHER1

ASG-AETHER1 Capacity overview

arn:aws:autoscaling:us-east-1:238256783131:autoScalingGroup:24e9118d-5bed-470c-9037-52d76dab39cb:autoScalingGroupName/ASG-AETHER1

Desired capacity

Scaling limits (Min - Max)

Desired capacity type

Status

Date created

Wed Mar 19 2025 14:37:35 GMT-0300 (Horário Padrão de Brasília)

Details

Integrations - new

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Launch template

Launch template

AMI ID

Instance type

Owner

Version

Security groups

Security group IDs

Create time

Launch template

AMI ID

Instance type

Owner

Version

Security groups

Security group IDs

Create time

EC2 > Auto Scaling groups > ASG-AETHER1

Elastic Block Store

Network & Security

Load Balancing

Auto Scaling

Settings

Details

Integrations - new

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Launch template

Launch template

AMI ID

Instance type

Owner

Version

Security groups

Security group IDs

Create time

Launch template

AMI ID

Instance type

Owner

Version

Security groups

Security group IDs

Create time

Network

Availability Zones

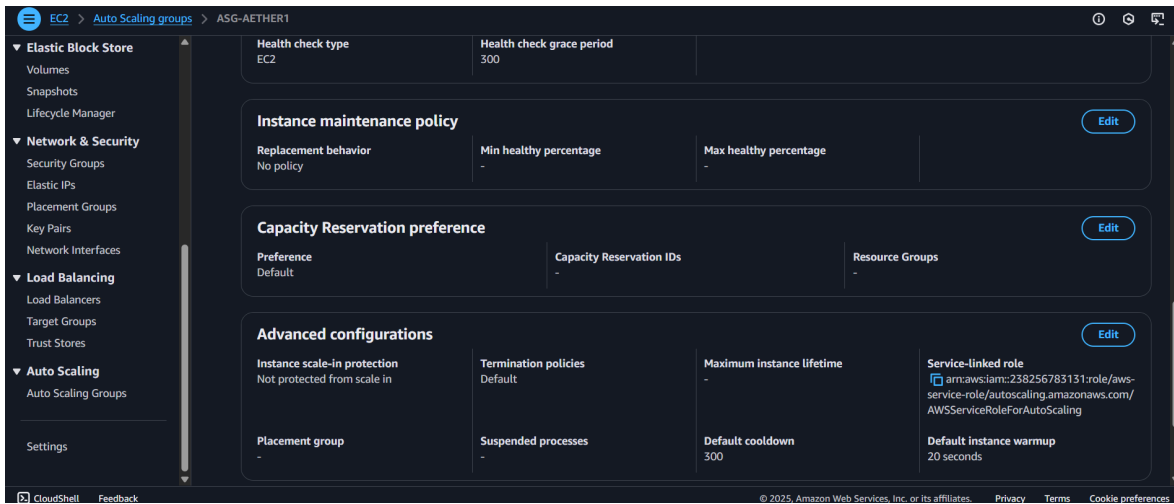
Subnet ID

Availability Zone distribution

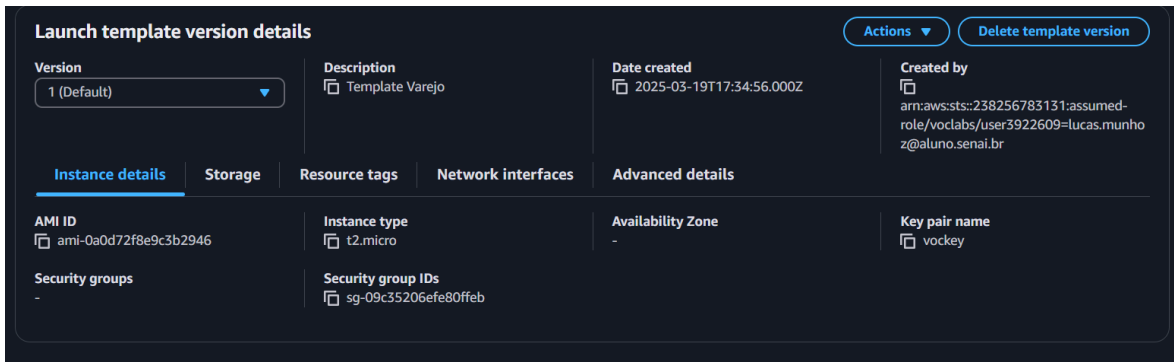
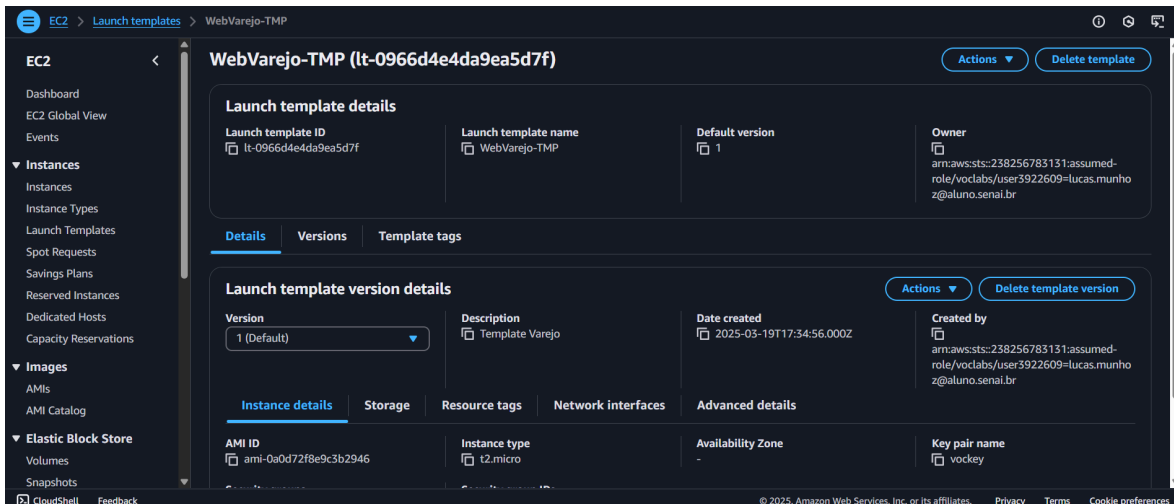
Availability Zones

Subnet ID

Availability Zone distribution

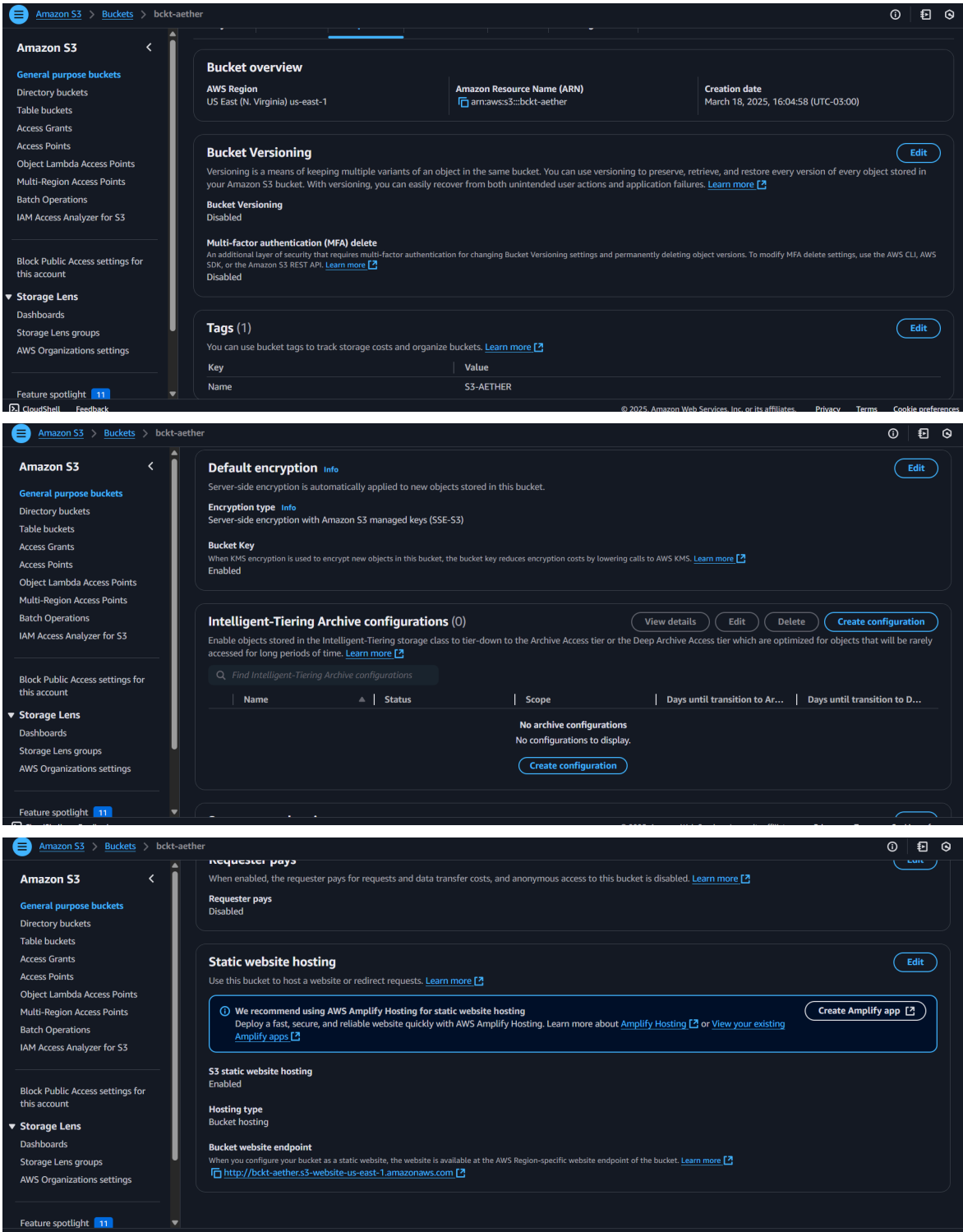


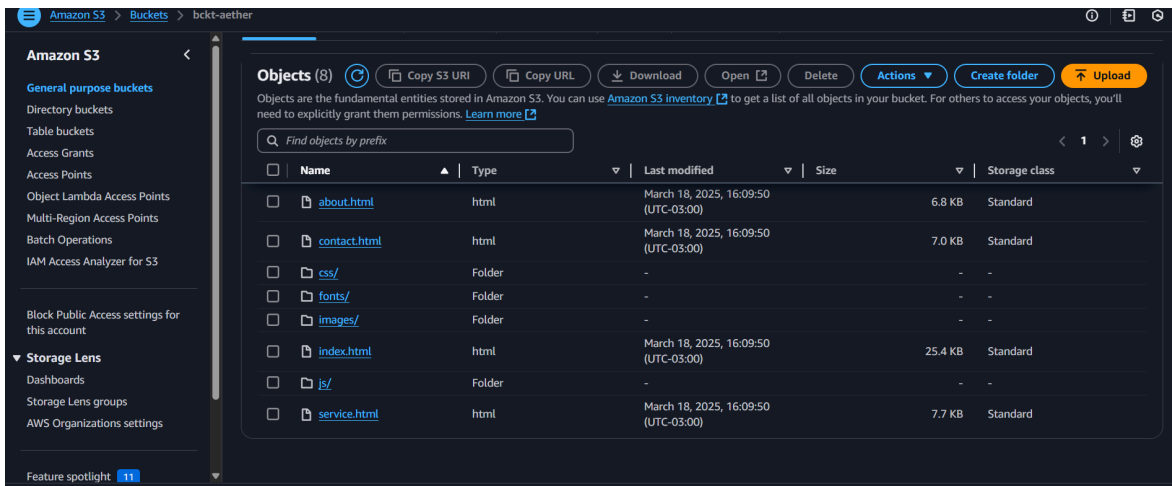
Templates usados em EC2 permite criar instâncias de forma padronizada e automatizada



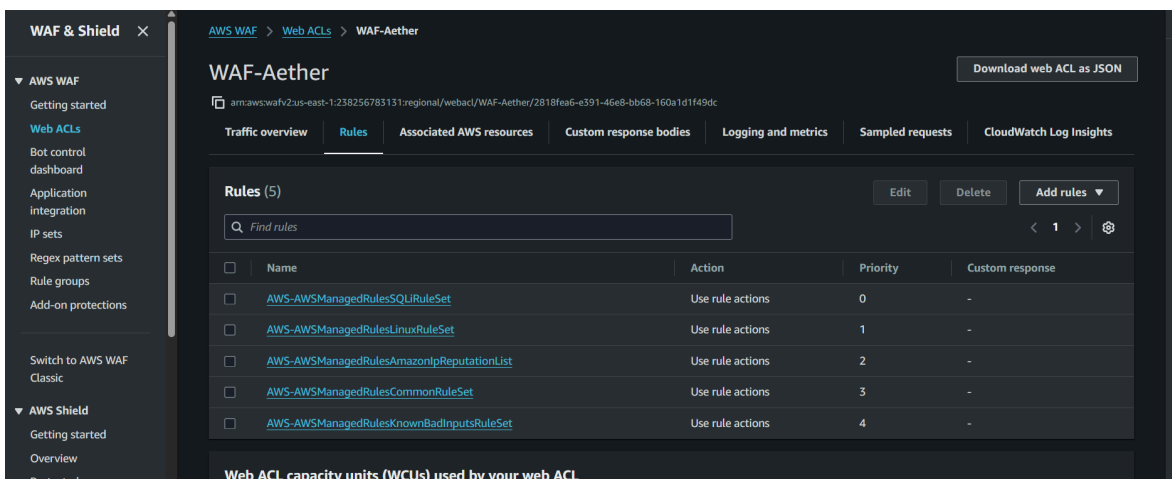
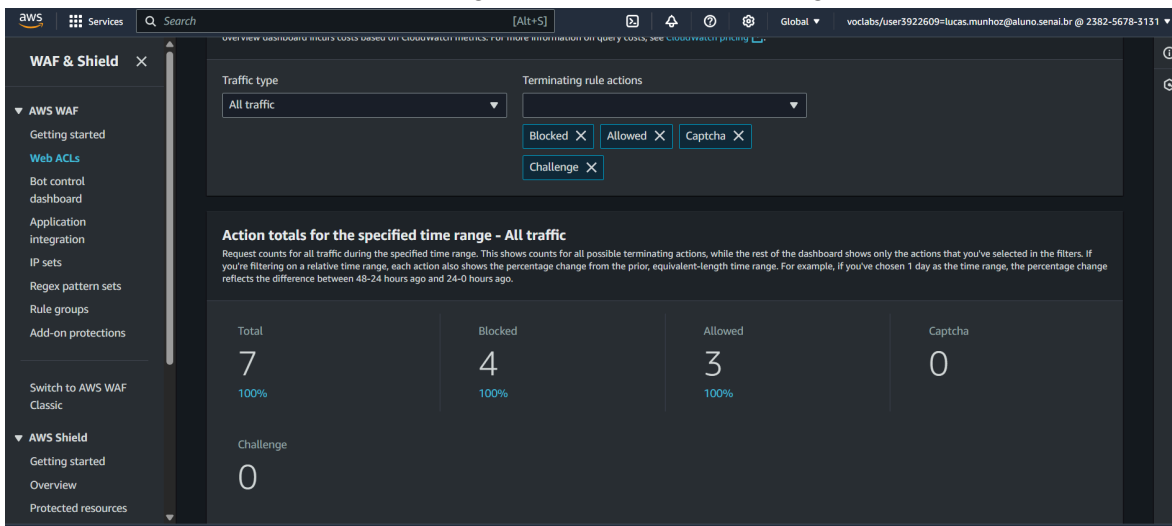
(OBS: Criamos uma AMI personalizada para já ter todas as configurações necessárias - httpd e cloudwatch agent)

**Serviço s3:** Serviço de armazenamento de objetos escalável, seguro e durável da AWS, usado para armazenar e recuperar qualquer quantidade de dados na nuvem.

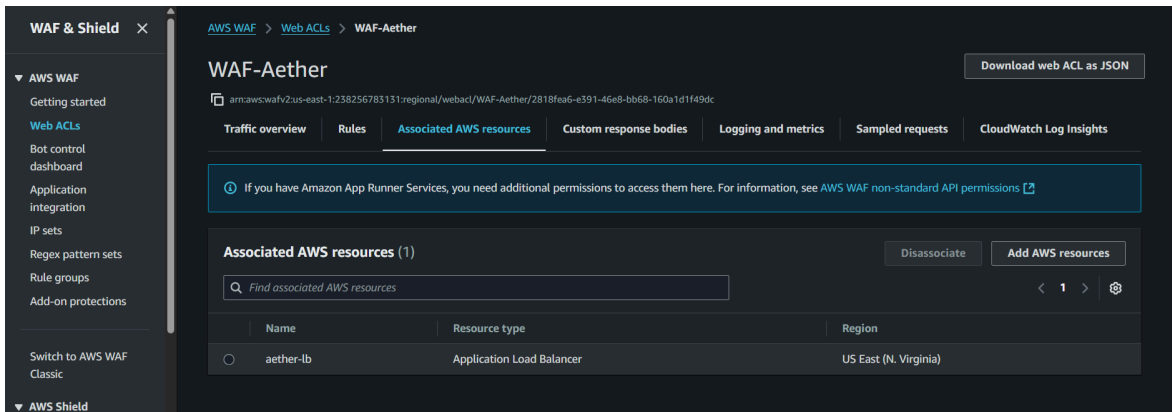




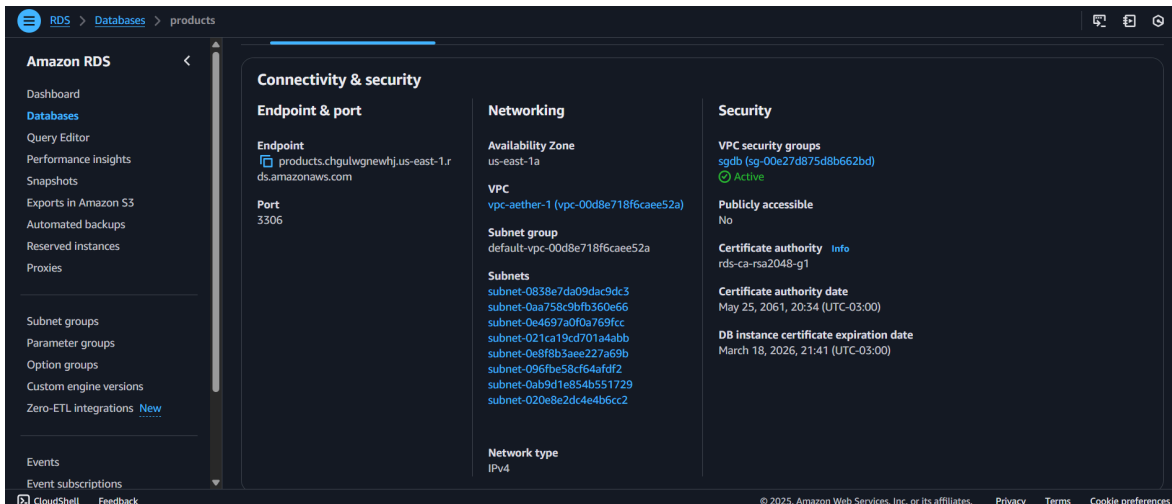
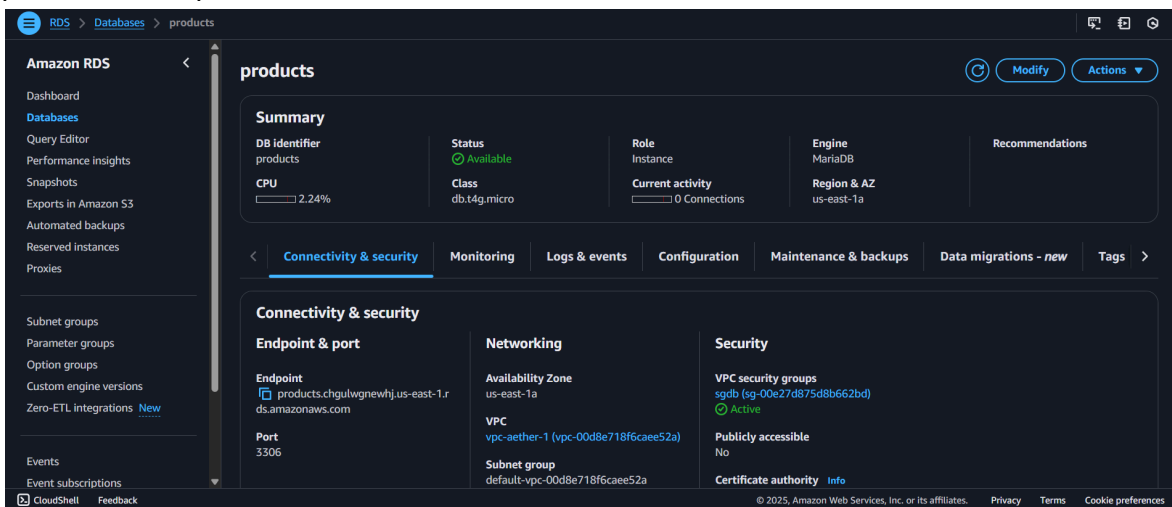
**WAF:** Firewall para aplicações web que protege contra ameaças como SQL Injection, XSS e ataques DDoS, filtrando tráfego malicioso antes de chegar aos recursos da AWS.







**Amazon Relational Database** é um serviço da Amazon Web Services (AWS) que permite criar, operar e escalar bancos de dados relacionais na nuvem



Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

New

Events

Event subscriptions

Security group rules (2)

Filter by Security group rules

| Security group              | Type               | Rule             |
|-----------------------------|--------------------|------------------|
| sgdb (sg-00e27d875d8b662bd) | CIDR/IP - Inbound  | 187.3.148.160/32 |
| sgdb (sg-00e27d875d8b662bd) | CIDR/IP - Outbound | 0.0.0.0/0        |

Replication (1)

Filter by Replication

| DB identifier | Role     | Region & AZ | Replication source | Replication state | Lag |
|---------------|----------|-------------|--------------------|-------------------|-----|
| products      | Instance | us-east-1a  | -                  | -                 | -   |

Amazon RDS

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

New

Events

Event subscriptions

CloudWatch (25)

Period 5 minutes

Add instance to compare

Monitoring

Search by metric

Alarm recommendations

3h 1d 1w

UTC timezone

BinLogDiskUsage

Bytes

15.0k

10.0k

5.0k

0

16.00 17.00 18.00

products

BurstBalance

Percent

100

80

60

40

20

0

16.00 17.00 18.00

products

CPUCreditBalance

Count

150

100

50

0

16.00 17.00 18.00

products

CPUCreditUsage

Count

CPUSurplusCreditBalance

Count

CPUSurplusCreditsCharged

Count

RDS

Databases

products

Amazon RDS

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

New

Events

Event subscriptions

Connectivity & security

Monitoring

Logs & events

Configuration

Maintenance & backups

Data migrations - new

Tags

Instance

Configuration

DB instance ID

products

Engine version

11.4.4

DB name

-

License model

General Public License

Option groups

default:manadb-11-4

In sync

Amazon Resource Name (ARN)

arn:aws:rds:us-east-1:23825678313:1:db:products

Resource ID

Instance class

Instance class

db.t4g.micro

vCPU

2

RAM

1 GB

Availability

Master username

admin

Master password

\*\*\*\*\*

IAM DB authentication

Not enabled

Storage

Encryption

Enabled

AWS KMS key

aws/rds

Storage type

General Purpose SSD (gp2)

Storage

20 GiB

Provisioned IOPS

-

Storage throughput

-

Storage autoscaling

Enabled

Monitoring

Monitoring type

Database Insights - Standard

Performance Insights

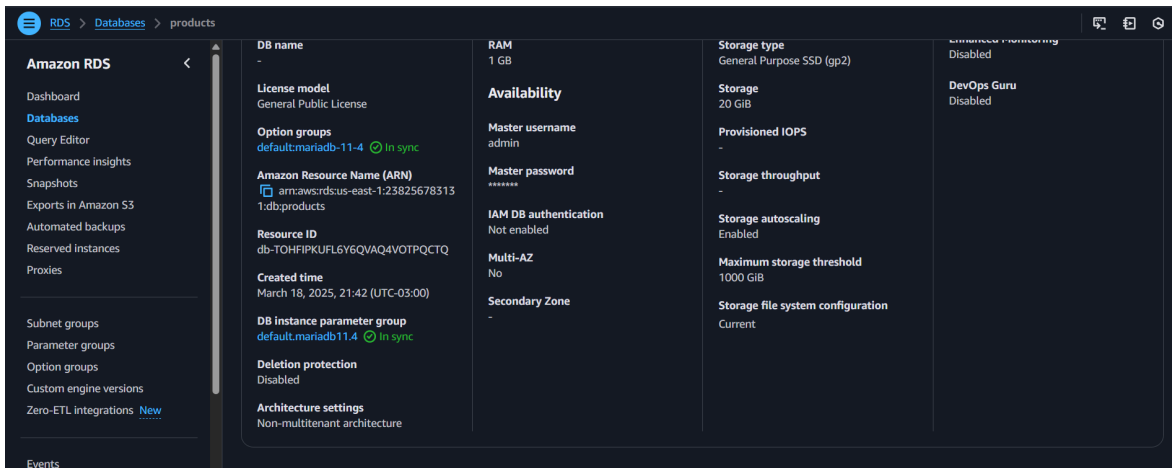
Disabled

Enhanced Monitoring

Disabled

DevOps Guru

Disabled



Mariadb é um sistema de gerenciamento de banco de dados relacional de código aberto. Ele permite armazenar e organizar dados em tabelas, colunas e linhas.

```
MariaDB [(none)]> USE products;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [products]> SHOW TABLES;
+-----+
| Tables_in_products |
+-----+
| fornecedores        |
| produtos            |
+-----+
2 rows in set (0.000 sec)

MariaDB [products]> SELECT * FROM fornecedores;
+----+-----+-----+-----+-----+
| id | nome_fornecedor | cnpj | endereco | cep |
+----+-----+-----+-----+-----+
| 1 | Investidora XD | 12389176203 | Rua Fulano Bairro Ciclano Cidade X | 1928370 |
| 2 | Betalino Innovations | 1298371239 | Rua Beltrano Bairro Fulano Cidade Y | 89761 |
| 3 | SigMat | 120837123 | Rua Sigma Bairro Alpha Cidade Omega | 9812376 |
| 4 | AlphaLino Metavisions | 198123756 | Rua Glenn Bairro Quag Cidade Mire | 9178263 |
| 5 | LOL Pereciveis | 123610283312 | Rua Yasuo Bairro Courtesy Cidade Combo | 8127378 |
+----+-----+-----+-----+-----+
5 rows in set (0.000 sec)
```

```
Enter password:
ERROR 1045 (28000): Access denied for user 'admin'@'localhost' (using password: YES)
ubuntu@ip-10-0-8-21:~$ mariadb -u admin -p
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 32
Server version: 10.11.8-MariaDB-Ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE products;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [products]> SHOW TABLES;
+-----+
| Tables_in_products |
+-----+
| fornecedores        |
| produtos            |
+-----+
2 rows in set (0.000 sec)

MariaDB [products]> []
```

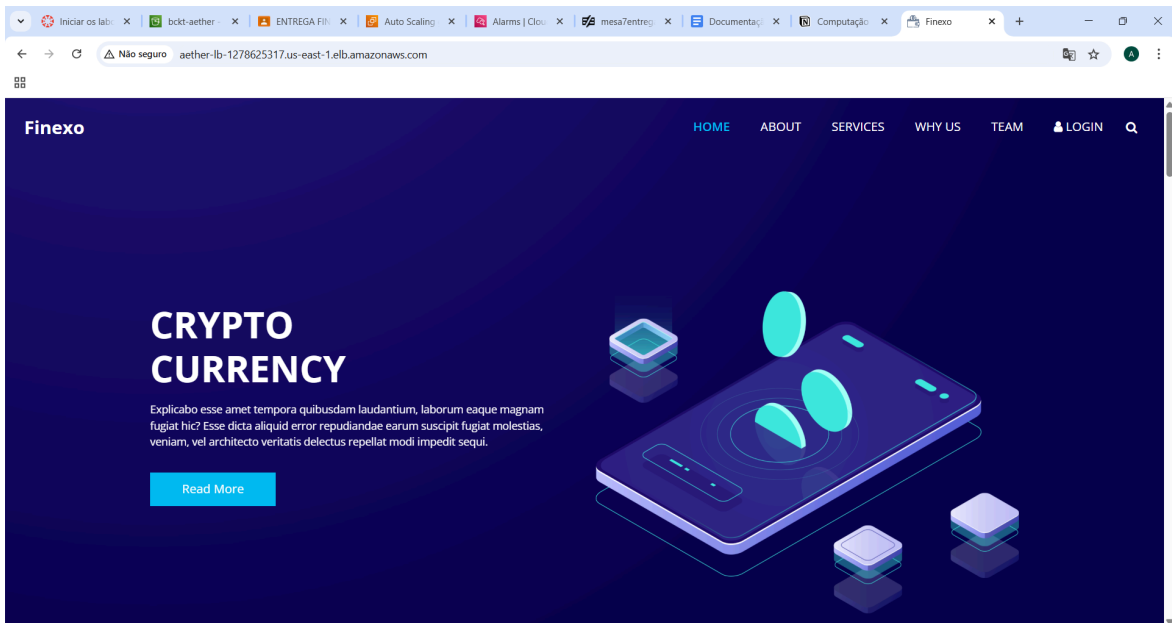
```
2 rows in set (0.000 sec)

MariaDB [products]> SELECT * FROM fornecedores;
+----+-----+-----+-----+-----+
| id | nome_fornecedor | cnpj | endereco | cep |
+----+-----+-----+-----+-----+
| 1 | Investidora XD | 12389176203 | Rua Fulano Bairro Ciclano Cidade X | 1928370 |
| 2 | Betalino Innovations | 1298371239 | Rua Beltrano Bairro Fulano Cidade Y | 89761 |
| 3 | SigMat | 120837123 | Rua Sigma Bairro Alpha Cidade Omega | 9812376 |
| 4 | AlphaLino Metavisions | 198123756 | Rua Glenn Bairro Quag Cidade Mire | 9178263 |
| 5 | LOL Pereciveis | 123610283312 | Rua Yasuo Bairro Courtesy Cidade Combo | 8127378 |
+----+-----+-----+-----+-----+
5 rows in set (0.000 sec)

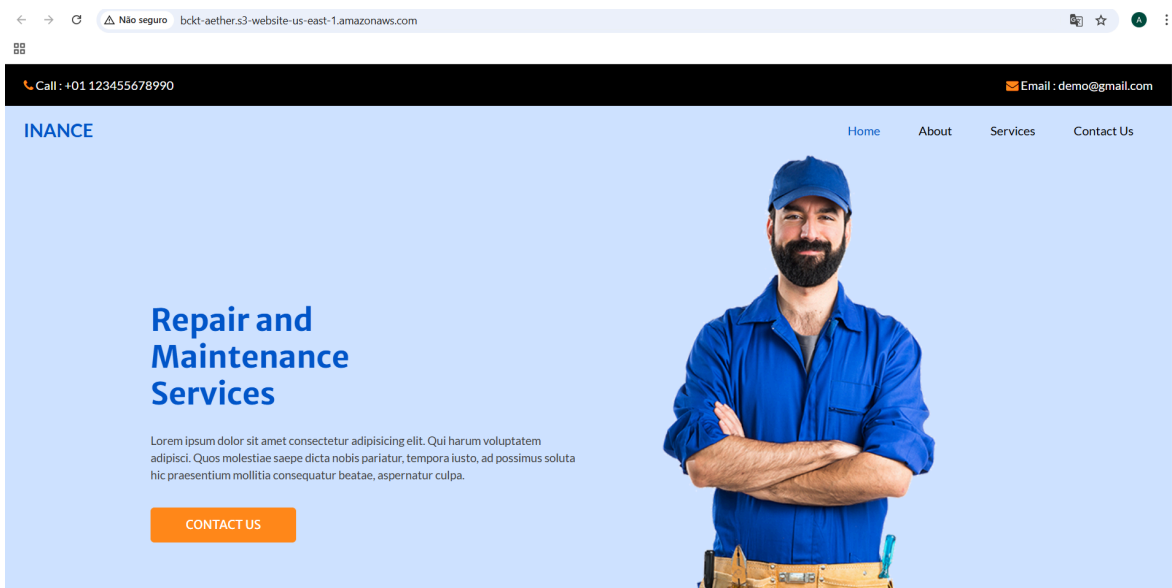
MariaDB [products]> SELECT * FROM produtos;
+----+-----+-----+-----+-----+
| id | nome_produto | valor | qtd_estoque | fornecedor_resp |
+----+-----+-----+-----+-----+
| 1 | Celular | R$700.000 | 500 | Investidora XD |
| 2 | Computador Gamer | R$800.000 | 100 | Betalino Innovations |
| 3 | Notebook | R$600.000 | 200 | SigMat |
| 4 | Oculos VR | R$650.000 | 350 | AlphaLino Metavisions |
| 5 | Arroz | R$400.000 | 11000 | LOL Pereciveis |
+----+-----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [products]> []
```

Print do site de varejo:



## Print do Site Static S3:



Topologia Aether:

